

Now denote by $\mathcal{V} \subset \mathbb{Z}_q^K$ the τK -separated subset of size $A_q(K, \tau K)$. For any distinct $v, v' \in \mathcal{V}$, by Lemma E.2 we have

$$|E_s(v_{\#}p^\varepsilon) \cap E_s(v'_{\#}p^\varepsilon)| \leq n_{\text{out}}d_{\text{out}}(K - \tau K) = n_{\text{out}}d_{\text{out}} \left(\frac{\log q}{Cq} \right) K.$$

On the other hand, it holds that

$$|\mathcal{M}_{v_{\#}p^\varepsilon}(X_{\text{in}}, X_{\text{out}}) \cap E_s(v_{\#}p^\varepsilon)| = |\mathcal{M}_{p^\varepsilon}(v^{-1}(X_{\text{in}}), v^{-1}(X_{\text{out}})) \cap E_s(p^\varepsilon)| \geq c \log K$$

due to Assumption 3, and similarly for v' . Since $M \geq cK$, we may choose C a large enough constant so that

$$C > \frac{n_{\text{out}}d_{\text{out}}}{c} \cdot \frac{K \log q}{q \log K}.$$

Then since

$$|\mathcal{M}_{v_{\#}p^\varepsilon}(X_{\text{in}}, X_{\text{out}}) \cap E_s(v_{\#}p^\varepsilon)|, |\mathcal{M}_{v'_{\#}p^\varepsilon}(X_{\text{in}}, X_{\text{out}}) \cap E_s(v'_{\#}p^\varepsilon)| > |E_s(v_{\#}p^\varepsilon) \cap E_s(v'_{\#}p^\varepsilon)|$$

it must hold that

$$\mathcal{M}_{v_{\#}p^\varepsilon}(X_{\text{in}}, X_{\text{out}}) \cap E_s(v_{\#}p^\varepsilon) \neq \mathcal{M}_{v'_{\#}p^\varepsilon}(X_{\text{in}}, X_{\text{out}}) \cap E_s(v'_{\#}p^\varepsilon).$$

Without loss of generality, we may suppose there exists a sparse edge $\xi = (X_{t-1}, X_t)$ of $v_{\#}p^\varepsilon$ that is included in the left-hand side of the above but not the right. Since $\mathcal{A}(\xi)$ is included in the computation of $r_{\mathcal{A}, v_{\#}p^\varepsilon}$ along $\mathcal{M}_{v_{\#}p^\varepsilon}(X_{\text{in}}, X_{\text{out}})$ but not of $r_{\mathcal{A}, v'_{\#}p^\varepsilon}$ along $\mathcal{M}_{v'_{\#}p^\varepsilon}(X_{\text{in}}, X_{\text{out}})$, denoting $\mathcal{A}^- := \mathcal{A}|_{S \times S \setminus \{\xi\}}$, it follows that

$$\begin{aligned} & \langle h_{v_{\#}p^\varepsilon}, h_{v'_{\#}p^\varepsilon} \rangle_{\mathcal{H}} \\ &= \mathbb{E}_{X_{\text{in}}, X_{\text{out}}, \mathcal{A}} \left[h_{v_{\#}p^\varepsilon}(X_{\text{in}}, X_{\text{out}}, \mathcal{A}) h_{v'_{\#}p^\varepsilon}(X_{\text{in}}, X_{\text{out}}, \mathcal{A}) \right] \\ &= \mathbb{E}_{X_{\text{in}}, X_{\text{out}}, \mathcal{A}} \left[\phi \circ r_{\mathcal{A}, v_{\#}p^\varepsilon}(\mathcal{M}_{v_{\#}p^\varepsilon}(X_{\text{in}}, X_{\text{out}})) \phi \circ r_{\mathcal{A}, v'_{\#}p^\varepsilon}(\mathcal{M}_{v'_{\#}p^\varepsilon}(X_{\text{in}}, X_{\text{out}})) \right] \\ &= \mathbb{E}_{X_{\text{in}}, X_{\text{out}}, \mathcal{A}^-} \left[\mathbb{E}_{\mathcal{A}(\xi)|\mathcal{A}^-} \left[\phi \circ r_{\mathcal{A}, v_{\#}p^\varepsilon}(\mathcal{M}_{v_{\#}p^\varepsilon}(X_{\text{in}}, X_{\text{out}})) \right] \phi \circ r_{\mathcal{A}, v'_{\#}p^\varepsilon}(\mathcal{M}_{v'_{\#}p^\varepsilon}(X_{\text{in}}, X_{\text{out}})) \right]. \end{aligned}$$

Adopting the shorthand $g1_A = g$ if A holds and $g1_A = e_G$ if not, we may write

$$r_{\mathcal{A}, v_{\#}p^\varepsilon}(\mathcal{M}_{v_{\#}p^\varepsilon}(X_{\text{in}}, X_{\text{out}})) = (\mathcal{A}_L \circ \mathcal{A}(X_{t-1}, X_t) \circ \mathcal{A}_R) \cdot \psi(X_{\text{in}})$$

where

$$\begin{aligned} \mathcal{A}_L &= \mathcal{A}(X_{T-1}, X_T) 1_{\{(X_{T-1}, X_T) \in E_s(v_{\#}p^\varepsilon)\}} \circ \cdots \circ \mathcal{A}(X_t, X_{t+1}) 1_{\{(X_t, X_{t+1}) \in E_s(v_{\#}p^\varepsilon)\}}, \\ \mathcal{A}_R &= \mathcal{A}(X_{t-2}, X_{t-1}) 1_{\{(X_{t-2}, X_{t-1}) \in E_s(v_{\#}p^\varepsilon)\}} \circ \cdots \circ \mathcal{A}(X_0, X_1) 1_{\{(X_0, X_1) \in E_s(v_{\#}p^\varepsilon)\}}. \end{aligned}$$

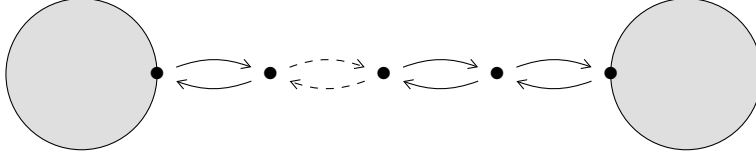


Figure 2: Sparse edge construction for the no-search scenario. The two circles represent the original dense clusters. Dashed edges have probability ε .

Since $\mathcal{A}(X_{t-1}, X_t)$ is uniformly distributed on G , the composition $\mathcal{A}_L \circ \mathcal{A}(X_{t-1}, X_t) \circ \mathcal{A}_R$ is also uniformly distributed on G conditioned on $\mathcal{A}^-$; note that this step relies crucially on the existence of left and right inverses in G . Hence

$$\mathbb{E}_{\mathcal{A}(\xi)|\mathcal{A}^-} [\phi \circ r_{\mathcal{A}, v_{\sharp} p^\varepsilon}(\mathcal{M}_{v_{\sharp} p^\varepsilon}(X_{\text{in}}, X_{\text{out}}))] = 0$$

for all $X_{\text{in}}, X_{\text{out}}, \mathcal{A}^-$, so that $\langle h_{v_{\sharp} p^\varepsilon}, h_{v'_{\sharp} p^\varepsilon} \rangle_{\mathcal{H}} = 0$. Thus the subset $\{h_{v_{\sharp} p^\varepsilon} : v \in \mathcal{V}\}$ of $\mathcal{H}$ is orthogonal with size at least

$$A_q(K, \tau K) \geq q^{\frac{\log \log q}{2Cq} K} \geq K^{\Omega(\log \log K)}$$

if $q = \Theta(K)$, which grows faster than any polynomial in K . In the case that $q/K \rightarrow \infty$, we may choose a smaller $q' = cK$ at the beginning to obtain the same lower bound. Therefore we have shown that $\text{SDA}(\mathcal{P}; \emptyset) \geq K^{\omega(1)}$.

Finally, if the minimum length in Assumption 3 is instead set to scale linearly as cK for some $0 < c < 1$, it suffices to set $C = 1$ and choose q (equivalently M) a large enough constant satisfying

$$n_{\text{out}} d_{\text{out}} \frac{\log q}{q} < c$$

to apply the same argument. Then the lower bound (21) becomes exponential in K .

Lower bounding $\text{SDA}(\mathcal{P}; \mathcal{M})$. We take an arbitrary kernel $p \in \mathbb{P}$ and make the following modification, pictured in Figure 2. Each sparse edge (x, y) is replaced by a set of states $z_0 = x, z_1, \dots, z_{q-1}, z_q = y$ such that each neighboring pair z_t, z_{t+1} for $t \in \mathbb{Z}_q$ are connected to each other via bidirectional edges of probability $O(1)$. However, one specific pair z_t, z_{t+1} is to be connected to each other with probability ε . Denote this index by v_j where $j \in [J]$ numbers the set of directly connected clusters; it holds that $J = |E_s| = \Theta(K)$ by Assumption 2. The points z_t for $t \leq v_j$ and for $t > v_j$ are appended to the clusters containing x and y , respectively. Since this is a bounded number of points all connected by constant probability edges, the extended clusters are still rapidly mixing. Then each vector $v = (v_j) \in \mathbb{Z}_q^J$ determines a kernel $p_v \in \mathcal{P}$, and

$$|E_s(p_v) \cap E_s(p_{v'})| = 2(J - d_H(v, v'))$$

holds for all $v, v' \in \mathbb{Z}_q^J$. We now repeat the argument from before to show orthogonality: by Lemma E.3 there exists a τJ -separated subset $\mathcal{V} \subset \mathbb{Z}_q^J$ of size $A_q(J, \tau J)$ for $\tau = 1 - \log q/q$,

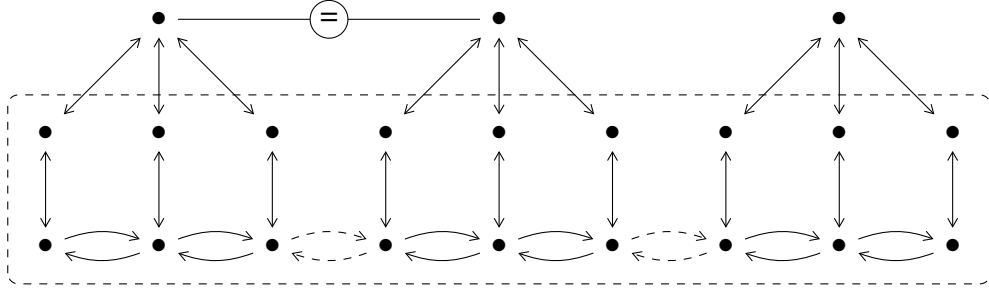


Figure 3: Graph construction for the local search scenario ($r = 2$). Dashed edges have probability ε . A local neighborhood of maximum distance one from the original graph is shown in the dashed box, which the learner is assumed to have full access to.

and by choosing q large enough we can ensure

$$|\mathcal{M}_{p_v}(X_{\text{in}}, X_{\text{out}}) \cap E_s(p_v)| \geq cK > \left(\frac{2 \log q}{q} \right) J \geq |E_s(p_v) \cap E_s(p_{v'})|.$$

The key aspect of this construction is that the support of p_v does not depend on v , as only the position of the sparse edge among q candidates change with v . This implies that querying the path $\mathcal{M}_{p_v}(X_{\text{in}}, X_{\text{out}})$ (and indeed the entire set of edges) does not reveal any information about the ground truth v , and $\{p_v : v \in \mathcal{V}\}$ satisfies the conditions to compute the SQ dimension with access to $\mathcal{M}$. Hence we have shown that

$$\text{SDA}(\mathcal{P}; \mathcal{M}) \geq A_q(J, \tau J) \geq e^{\Omega(K)}.$$

Lower bounding $\text{SDA}(\mathcal{P}; \text{nb}(\mathcal{M}))$. We construct the graph depicted in Figure 3 as follows. We start with nK clusters of size M laid out in side by side and connect neighboring clusters with bidirectional edges of probability ε . From all states extend a ‘rod’ of probability $O(1)$ bidirectional edges of bounded length r (the vertically arranged states, here $r = 2$), similarly to Figure 2. Join the endpoints of all rods originating from each cluster into a single ‘endpoint’ state. For $\mathcal{D}$, we assume that $(X_{\text{in}}, X_{\text{out}})$ are sampled only from the original clusters (bottom horizontal line of states) and $\mathcal{M}_p$ only returns paths along this line.

Choose a size K subset B of the low probability edges to be sparse edges, viewed as a subset of $[nK]$. For each of the edges (x, y) not in B , identify the endpoint states of the clusters containing x, y . The identified clusters will merge into a single larger rapidly mixing cluster, so that (x, y) is indeed no longer a sparse edge. Denote the resulting kernel as p_B . We have the following intersection constraint bound:

Lemma E.5. *For any $n \geq 5$, there exists a size $e^{\Omega(K)}$ set $\mathcal{B}$ of size K subsets of $[nK]$ such that $|B \cap B'| \leq cK$ for all $B, B' \in \mathcal{B}$.*